Lecture 4

Simple, solvable and nilpotent groups

Objectives. *Lecture 3 found orders, such as pq and 12, at which no group is simple. This lecture proves that A_5 is simple, and shows how every finite group is built from simple groups: a composition series breaks it into simple factors, and by the Jordan–Hölder theorem the factors do not depend on the series. The groups built from abelian pieces are the solvable groups; S_4 is one, and S_n for $n \geq 5$ is not. We also prove that p-groups are nilpotent and nilpotent groups are solvable. The commutator subgroup goes on to Lecture 6, and solvability to Lecture 9.*

By the end of this lecture you will be able to:

- Decide whether a class of S_n inside A_n is one class of A_n or two, and prove that A_5 is simple.
- Prove that every finite group has a composition series, and that any two have the same factors, up to order and isomorphism.
- Compute the commutator subgroup and the derived series of a small group, and prove that S_n is solvable exactly when $n \leq 4$.
- Prove that p-groups are nilpotent and nilpotent groups solvable, and show that S_3 is solvable but not nilpotent.

4.1 Simple groups; A_5

A group of prime order is simple (Definition 3.13). To find a non-abelian simple group we need the classes of A_5, since a normal subgroup is a union of classes (Proposition 2.2). The classes of S_n are its cycle types (Theorem 2.16). A class contained in A_n can split when conjugation is restricted to even permutations.

We saw this in Example 3.20: the eight 3-cycles form one class in S_4, but two classes of four in A_4. The next lemma gives a test for when this happens.

Lemma 4.1 (Classes of A_n). *Let $n \geq 2$, let $\sigma \in A_n$, and let $[\sigma]$ be its class in S_n. Then $[\sigma] \subseteq A_n$, and:*

(i) if $C_{S_n}(\sigma)$ contains an odd permutation, $[\sigma]$ is one class of A_n;

(ii) if $C_{S_n}(\sigma) \leq A_n$, $[\sigma]$ is the union of two classes of A_n, each with $|[\sigma]|/2$ elements.

Proof. The sign is a homomorphism to an abelian group, so

$$\operatorname{sgn}(\tau\sigma\tau^{-1}) = \operatorname{sgn}\sigma = 1.$$

Thus $[\sigma] \subseteq A_n$.

Write $C = C_{S_n}(\sigma)$. The centraliser of σ in A_n is $C \cap A_n$, so by the class-size formula (Theorem 1.20) the class of σ in A_n has

$$\frac{|A_n|}{|C \cap A_n|} = \frac{n!/2}{|C \cap A_n|}$$

elements. It lies in $[\sigma]$, which has $n!/|C|$ elements.

(i) The sign restricted to C is onto $\{\pm 1\}$, with kernel $C \cap A_n$; so $|C \cap A_n| = |C|/2$ by the first isomorphism theorem (Theorem 2.5), and the class of σ in A_n has $n!/|C|$ elements. So it is all of $[\sigma]$.

(ii) Now $C \cap A_n = C$, so the class of σ in A_n has $|[\sigma]|/2$ elements. Let $\sigma' = (1\,2)\,\sigma\,(1\,2)^{-1}$, which exists as $n \geq 2$.

Two classes cover $[\sigma]$. An element of $[\sigma]$ is $\tau\sigma\tau^{-1}$ with $\tau \in S_n$. If τ is even, it lies in the class of σ in A_n. If τ is odd, then $\rho = \tau(1\,2)$ is even, $\tau = \rho\,(1\,2)$, and

$$\begin{aligned}\tau\sigma\tau^{-1} &= \rho\,(1\,2)\,\sigma\,(1\,2)^{-1}\rho^{-1}\\ &= \rho\,\sigma'\,\rho^{-1},\end{aligned}$$

which lies in the class of σ' in A_n.

The halves. The class of σ holds only half of $[\sigma]$, so the two classes differ; classes of A_n that differ are disjoint, so the class of σ' is the other half. ∎

Example 4.2 (The classes of A_5)**.** The even cycle types in S_5 are (1^5), $(2^2,1)$, $(3,1^2)$ and (5) (Theorem 2.19). For each, Lemma 4.1 asks for an odd permutation that commutes with a representative:

type	σ	$\lvert[\sigma]\rvert = 120/z_\mu$	odd, commuting	A_5-class sizes
(1^5)	e	1	$(1\,2)$	1
$(2^2,1)$	$(1\,2)(3\,4)$	15	$(1\,2)$	15
$(3,1^2)$	$(1\,2\,3)$	20	$(4\,5)$	20
(5)	$(1\,2\,3\,4\,5)$	24	none	12, 12

For example, a 3-cycle has $z_{(3,1^2)} = 3 \cdot 2! = 6$, so its S_5-class has $120/6 = 20$ elements. The transposition $(4\,5)$ is odd and commutes with $(1\,2\,3)$, so these 20 elements form one A_5-class. In the last row the centraliser has order $z_{(5)} = 5$, so it is $\langle(1\,2\,3\,4\,5)\rangle \leq A_5$. The class of five-cycles therefore splits into two classes of 12. These classes exhaust A_5: their sizes sum to $1 + 15 + 20 + 12 + 12 = 60 = |A_5|$.

Example 4.3 (Guard: a class of S_n need not be a class of A_n)**.** The 24 five-cycles are one class of S_5 but two of A_5. The odd permutation $(1\,2)$ conjugates $(1\,2\,3\,4\,5)$ to

$$(1\,2)(1\,2\,3\,4\,5)(1\,2)^{-1} = (2\,1\,3\,4\,5),$$

and no even permutation does. Indeed, if an even τ had the same effect, then $(1\,2)^{-1}\tau$ would be an odd element of the centraliser $\langle(1\,2\,3\,4\,5)\rangle \leq A_5$, a contradiction. Counted by cycle type, A_5 would have four classes, not five; Lecture 7 counts the irreducible representations of a group by its classes.

Theorem 4.4 (A_5 is simple)**.** *The alternating group A_5 is simple.*

Proof. Let $N \trianglelefteq A_5$. Then N is a union of classes of A_5 containing $\{e\}$ (Proposition 2.2), and $|N|$ divides 60 (Lagrange). By Example 4.2, the sums of 1 and some of 15, 20, 12, 12 are

$$1,\ 13,\ 16,\ 21,\ 25,\ 28,\ 33,\ 36,\ 40,\ 45,\ 48,\ 60,$$

and of these only 1 and 60 divide 60. So $N = \mathbf{1}$ or $N = A_5$; and $A_5 \neq \mathbf{1}$. ∎

Example 4.5 (A_5 is not abelian)**.** Two three-cycles need not commute:

$$(1\,2\,3)(3\,4\,5) = (1\,2\,3\,4\,5),$$
$$(3\,4\,5)(1\,2\,3) = (1\,2\,4\,5\,3).$$

The two products differ, so A_5 is not abelian.

4.2 Composition series and Jordan–Hölder

A non-trivial proper normal subgroup N of a finite group G gives two groups of smaller order, N and G/N (§ 3.4). A composition series records a sequence of such quotients until each factor is simple. To compare two series, we first prove the second isomorphism theorem.

Theorem 4.6 (Second isomorphism theorem)**.** *Let $H \leq G$ and $N \trianglelefteq G$. Then $HN = \{hn : h \in H,\ n \in N\}$ is a subgroup of G with $N \trianglelefteq HN$; $H \cap N \trianglelefteq H$; and*

$$H/(H \cap N) \cong HN/N, \qquad h(H \cap N) \mapsto hN.$$

Proof. *HN is a subgroup.* It contains e. For $h, h' \in H$ and $n, n' \in N$,

$$(hn)(h'n') = (hh')\,(h'^{-1}nh')\,n',$$
$$(hn)^{-1} = h^{-1}\,(hn^{-1}h^{-1}),$$

which lie in HN, as N is normal in G, hence in HN.

The isomorphism. The quotient map restricted to H, $h \mapsto hN$, is a homomorphism $H \to HN/N$. It is onto, since $(hn)N = hN$, and its kernel is $H \cap N$, normal in H as a kernel. So the first isomorphism theorem (Theorem 2.5) applies. ∎

For the proof of Jordan–Hölder we also need products of normal subgroups. If M and N are both normal in G, then MN is a subgroup by the second isomorphism theorem. It is normal too: for $g \in G$,

$$g\,MN\,g^{-1} = (gMg^{-1})(gNg^{-1}) = MN.$$

Definition 4.7 (Composition series)**.** A *composition series* of G is a chain

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \dots \trianglerighteq G_r = \mathbf{1}$$

in which each G_{i+1} is a proper normal subgroup of G_i and each *factor* G_i/G_{i+1} is simple; r is its *length*. Two composition series are *equivalent* if they have the same length and, after reordering, their corresponding factors are isomorphic.

Each G_{i+1} need be normal only in G_i (Example 2.22). The trivial group has one composition series, $\mathbf{1}$, of length 0.

For example, the normal subgroups from § 2.1 give

$$S_3 \trianglerighteq A_3 \trianglerighteq \mathbf{1}.$$

Its factors have orders 2 and 3, so they are simple. This is a composition series, with factors isomorphic to C_2 and C_3.

Definition 4.8 (Maximal normal subgroup)**.** A proper normal subgroup N of G is *maximal normal* if no normal subgroup K of G has $N < K < G$.

Lemma 4.9. *For $N \trianglelefteq G$, the quotient G/N is simple if and only if N is a maximal normal subgroup of G.*

Proof. By the correspondence theorem (Theorem 2.5), the normal subgroups of G/N are the K/N with $N \leq K \trianglelefteq G$. The trivial subgroup and the whole quotient correspond to N and G, respectively. Also $G/N \neq \mathbf{1}$ exactly when $N \neq G$. Thus the quotient is simple exactly when N is proper and no normal subgroup lies strictly between N and G. ∎

Theorem 4.10 (Jordan–Hölder). *Let G be a finite group.*

(i) G has a composition series.

(ii) Any two composition series of G are equivalent.

Proof. (i) By induction on $|G|$.

The base. For $G = \mathbf{1}$, take $\mathbf{1}$.

The top factor. If $G \neq \mathbf{1}$, let N be a proper normal subgroup of largest order ($\mathbf{1}$ is one). It is maximal normal, so G/N is simple (Lemma 4.9).

The tail. By induction N has a composition series, and

$$G \trianglerighteq N \trianglerighteq \ldots \trianglerighteq \mathbf{1}$$

is one for G.

(ii) By induction on $|G|$; for $G = \mathbf{1}$ there is only one series. Let $G \neq \mathbf{1}$, with two composition series

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \ldots \trianglerighteq G_r = \mathbf{1},$$
$$G = H_0 \trianglerighteq H_1 \trianglerighteq \ldots \trianglerighteq H_s = \mathbf{1},$$

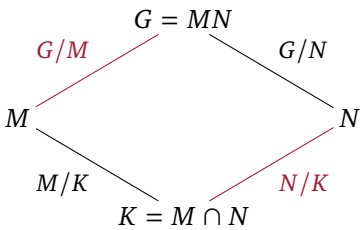
and let $M = G_1$, $N = H_1$, maximal normal by Lemma 4.9.

Case $M = N$. The tails from G_1 and from H_1 are composition series of M, and $|M| < |G|$. By induction they are equivalent; add the common factor G/M.

Case $M \neq N$: the diamond. Then $N \not\leq M$, for otherwise $N < M < G$, against the maximality of N. So MN is normal in G and properly contains M; by the maximality of M, $MN = G$. Let $K = M \cap N$. By Theorem 4.6, K is normal in M and in N, and the two products MN and NM agree because M and N are normal. Thus

$$M/K \cong MN/N = G/N,$$
$$N/K \cong MN/M = G/M,$$

both simple: opposite sides of the diamond are isomorphic. In the diagram each edge is labelled by the quotient of the two groups it joins. Opposite edges carry isomorphic quotients; one pair is shown in the accent colour.



Case $M \neq N$: the induction. Take a composition series $K = K_0 \trianglerighteq \ldots \trianglerighteq K_t = \mathbf{1}$ of K, by existence. As M/K is simple,

$$M \trianglerighteq K_0 \trianglerighteq \ldots \trianglerighteq K_t = \mathbf{1}$$

is a composition series of M, and by induction it is equivalent to $G_1 \trianglerighteq \ldots \trianglerighteq G_r$. So the first series of G is equivalent to

$$G \trianglerighteq M \trianglerighteq K_0 \trianglerighteq \ldots \trianglerighteq K_t = \mathbf{1},$$

with factors isomorphic to G/M, G/N and those of K.

Similarly, N/K is simple, so adjoining the same series of K gives a composition series of N. By induction it is equivalent to the tail of the second series. Thus the second series has factors G/N, G/M and those of K: the same factors as the first, in another order. Thus the two series are equivalent. ∎

So the *composition factors* of G, with their multiplicities, are determined by G up to isomorphism.

Example 4.11 (Guard: the factors do not determine the group)**.** The cyclic group $C_6 = \langle g \rangle$ has two composition series,

$$C_6 \trianglerighteq \langle g^2 \rangle \trianglerighteq \mathbf{1}, \qquad C_6 \trianglerighteq \langle g^3 \rangle \trianglerighteq \mathbf{1},$$

with factors C_2, C_3 and C_3, C_2: the order of the factors depends on the series. In the diamond above, take $G = C_6$, $M = \langle g^2 \rangle$, $N = \langle g^3 \rangle$ and $K = \mathbf{1}$; the opposite factors are $M/K \cong G/N \cong C_3$ and $N/K \cong G/M \cong C_2$. The composition series of S_3 above has the same factors, though S_3 is not abelian. Thus the composition factors do not determine the group up to isomorphism. Even among abelian groups, C_4 and V_4 differ (only C_4 is cyclic), but both have factors C_2, C_2.

4.3 Solvable groups and the abelianisation

We now use commutators to measure the failure of elements to commute. They lead to a normal subgroup whose quotient is abelian.

Definition 4.12 (Commutators)**.** The *commutator* of $a, b \in G$ is

$$[a, b] = aba^{-1}b^{-1}.$$

The *commutator subgroup* G' is the subgroup generated by all commutators.[a]

[a] Dummit–Foote and Isaacs write $[x, y] = x^{-1}y^{-1}xy$: our $[x^{-1}, y^{-1}]$, a conjugate of $[x, y]$, not in general its inverse. Both generate G'.

So $[a, b] = e$ exactly when $ab = ba$, and $G' = \mathbf{1}$ exactly when G is abelian.

Proposition 4.13 (The abelianisation)**.**

(i) A homomorphism φ carries commutators to commutators, $\varphi([a, b]) = [\varphi(a), \varphi(b)]$; and $G' \trianglelefteq G$.

(ii) For $N \trianglelefteq G$, the quotient G/N is abelian if and only if $G' \leq N$.

(iii) G/G' is abelian, and every homomorphism φ from G to an abelian group factors through it: $\varphi = \bar{\varphi} \circ q$, with $q \colon G \to G/G'$ the quotient map.

Proof. (i) As φ is a homomorphism,

$$\varphi(aba^{-1}b^{-1}) = \varphi(a)\varphi(b)\varphi(a)^{-1}\varphi(b)^{-1}.$$

Conjugation by g is a homomorphism $G \to G$, so it carries commutators to commutators, and $gG'g^{-1} \subseteq G'$, enough for normality (§ 2.1).

(ii) Let $q \colon G \to G/N$ be the quotient map. Homomorphisms preserve commutators, so $q([a, b]) = [q(a), q(b)]$, which is the identity N exactly when $q(a)$ and $q(b)$ commute. So G/N is abelian exactly when every commutator lies in $N = \ker q$, equivalently $G' \leq N$.

(iii) The quotient criterion makes G/G' abelian. If φ maps to an abelian group, then

$$\varphi([a,b]) = [\varphi(a), \varphi(b)] = e,$$

so $G' \leq \ker \varphi$, and φ factors through G/G' by the factorisation theorem (Theorem 2.5). ∎

Definition 4.14 (Abelianisation). The quotient G/G' is the *abelianisation* of G.

The quotient criterion says that every abelian quotient of G is a quotient of G/G'. Factorisation and composition with the quotient map give a bijection between homomorphisms $G \to \mathbb{C}^\times$ and homomorphisms $G/G' \to \mathbb{C}^\times$. The former are the one-dimensional representations of G used in Lecture 6.

Example 4.15 (S_3'). As S_3/A_3 has order 2, it is abelian, and $S_3' \leq A_3$ by the quotient criterion. With $(1\,3)^{-1} = (1\,3)$ and $(1\,2)^{-1} = (1\,2)$,

$$\begin{aligned}[(1\,3),(1\,2)] &= (1\,3)(1\,2)(1\,3)(1\,2) \\ &= (1\,2\,3)(1\,2\,3) = (1\,3\,2),\end{aligned}$$

which generates A_3. So $S_3' = A_3$, and $A_3' = \mathbf{1}$, as A_3 is abelian.

Definition 4.16 (Solvable group). The *derived series* of G is $G^{(0)} = G$ and $G^{(i+1)} = \left(G^{(i)}\right)'$:

$$G = G^{(0)} \trianglerighteq G^{(1)} \trianglerighteq G^{(2)} \trianglerighteq \cdots.$$

The group G is *solvable* if $G^{(m)} = \mathbf{1}$ for some m.

Each term is normal in the one before, with abelian quotient (Proposition 4.13). Abelian groups are solvable, and so is S_3 (Example 4.15). A non-abelian simple group G is not solvable. Indeed, G' is a non-trivial normal subgroup, so $G' = G$ and $G^{(i)} = G$ for every i. In particular, A_5 is not solvable: it is simple and non-abelian (Theorem 4.4 and Example 4.5).

Theorem 4.17 (Solvable groups).

(i) If $H \leq G$, then $H^{(i)} \leq G^{(i)}$ for every i. So subgroups of solvable groups are solvable.

(ii) If $\varphi\colon G \to K$ is a surjective homomorphism, then $\varphi(G^{(i)}) = K^{(i)}$ for every i. So quotients of solvable groups are solvable.

(iii) If $N \trianglelefteq G$, and N and G/N are solvable, then G is solvable.

(iv) G is solvable if and only if it has a chain $G = H_0 \trianglerighteq H_1 \trianglerighteq \ldots \trianglerighteq H_s = \mathbf{1}$, each H_{i+1} normal in H_i, with every H_i/H_{i+1} abelian.

Proof. (i) By induction on i: the commutators of elements of $H^{(i)}$ are commutators of elements of $G^{(i)}$, so $(H^{(i)})' \leq (G^{(i)})'$.

(ii) By induction on i: φ maps the commutators of $G^{(i)}$ onto those of $\varphi(G^{(i)}) = K^{(i)}$ (Proposition 4.13), so it maps $G^{(i+1)}$ onto $K^{(i+1)}$.

(iii) Let $(G/N)^{(n)} = \mathbf{1}$ and $N^{(m)} = \mathbf{1}$. By the image formula for derived series, the quotient map sends $G^{(n)}$ onto $(G/N)^{(n)} = \mathbf{1}$. Its kernel is N, so $G^{(n)} \leq N$. By the subgroup inclusion for derived series,

$$G^{(n+m)} = \left(G^{(n)}\right)^{(m)} \leq N^{(m)} = \mathbf{1}.$$

(iv) The derived series of a solvable group is such a chain. Conversely, $G^{(i)} \leq H_i$ by induction: if $G^{(i)} \leq H_i$, use the subgroup inclusion and the abelian quotient criterion (Proposition 4.13) in H_i,

$$G^{(i+1)} \leq H_i' \leq H_{i+1}.$$

So $G^{(s)} \leq H_s = \mathbf{1}$. ∎

When $N \trianglelefteq G$, the group G is called an *extension* of G/N by N. The theorem says that an extension of a solvable group by a solvable group is solvable.

Corollary 4.18. *S_n is solvable if and only if $n \leq 4$.*

Proof. In $S_4 \trianglerighteq A_4 \trianglerighteq V_4 \trianglerighteq \mathbf{1}$ each term is normal in the one before (§ 2.6); the first two factors have prime order, so are cyclic (Corollary 1.17), and V_4 is abelian. By the chain criterion of Theorem 4.17, S_4 is solvable. For $n \leq 4$, S_n is the subgroup of S_4 fixing $n+1, \ldots, 4$; solvability passes to subgroups.

For $n \geq 5$, S_n contains A_5, as the even permutations of $\{1, \ldots, 5\}$. The group A_5 is not solvable, and solvability passes to subgroups. So S_n is not solvable. ∎

4.4 Nilpotent groups and p-groups

An abelian group is its own centre. To define nilpotence, start at the trivial subgroup and repeatedly take the preimage of the centre of the quotient.

Definition 4.19 (Nilpotent group). The *upper central series* of G is $Z_0 = \mathbf{1}$ and, for $i \geq 0$, the subgroup $Z_{i+1} \geq Z_i$ with

$$Z_{i+1}/Z_i = Z(G/Z_i).$$

The group G is *nilpotent* if $Z_m = G$ for some m.

The definition makes sense by induction on i. If $Z_i \trianglelefteq G$, let $q_i \colon G \to G/Z_i$ be the quotient map. The subgroup Z_{i+1} is the preimage under q_i of the centre of G/Z_i. Indeed, that centre is normal in G/Z_i, and by the correspondence theorem (Theorem 2.5) it is Z_{i+1}/Z_i for exactly one Z_{i+1}, again normal in G. So $Z_1 = Z(G)$, and abelian groups are nilpotent.

Example 4.20 (The group H_1 of order 8). In H_1 (Example 3.4), the first step is the centre,

$$Z_1 = Z(H_1) = \{e,\ (1\,2)(3\,4)\}.$$

The quotient H_1/Z_1 has order 4, so it is abelian (Theorem 3.3), and it is its own centre. So $Z_2 = H_1$:

$$\mathbf{1} = Z_0 < Z_1 < Z_2 = H_1.$$

The group H_1 is nilpotent, though not abelian.

Theorem 4.21 (Nilpotent groups).

(i) Every group of order p^a, p prime, is nilpotent.

(ii) Every nilpotent group is solvable.

Proof. (i) Let $|P| = p^a$. If $Z_i \neq P$, then P/Z_i is a non-trivial p-group (Lagrange), so its centre Z_{i+1}/Z_i is not trivial (Theorem 3.3), and

$$|Z_{i+1}| \geq p\ |Z_i|\,.$$

So, for each i, either $Z_i = P$ or $|Z_i| \geq p^i$. If $Z_a \neq P$, then $|Z_a| \geq p^a = |P|$, impossible for a proper subgroup. Thus $Z_a = P$.

(ii) Let $Z_m = G$. Each Z_i is normal in G, hence in Z_{i+1}, and each factor $Z_{i+1}/Z_i = Z(G/Z_i)$ is abelian. So the chain

$$G = Z_m \trianglerighteq Z_{m-1} \trianglerighteq \ldots \trianglerighteq Z_0 = \mathbf{1}$$

is a finite normal chain with abelian factors. The chain criterion in Theorem 4.17 shows that G is solvable. ∎

Example 4.22 (Guard: solvable is not nilpotent)**.** The group S_3 is solvable (Example 4.15), but $Z(S_3) = \mathbf{1}$ (Example 1.21). So $Z_1 = Z_0 = \mathbf{1}$, and as each Z_{i+1} depends only on Z_i, every Z_i is $\mathbf{1}$: S_3 is not nilpotent. Thus a solvable group need not be nilpotent.

4.5 Worked example

We find the composition series, derived series and abelianisation of S_4.

The composition series. The normal subgroups of S_4 are $\mathbf{1}$, V_4, A_4, S_4 (§ 2.6). So A_4 is the only maximal normal subgroup, and every composition series begins $S_4 \trianglerighteq A_4$ (Lemma 4.9).

A normal subgroup of A_4 is a union of its classes, of sizes 1, 3, 4, 4 (Example 3.20), containing $\{e\}$ and of order dividing 12. The possible orders are

$$\{1, 4, 12\}.$$

The only union of size 4 is $\{e\}$ with the class of size 3, namely V_4. So the normal subgroups of A_4 are $\mathbf{1}$, V_4, A_4. The next term is a maximal normal subgroup of A_4 (Lemma 4.9), so it is V_4. Every subgroup of the abelian group V_4 is normal. Its three subgroups of order 2 have index 2 and are maximal; the trivial subgroup is not maximal. So S_4 has exactly three composition series,

$$S_4 \trianglerighteq A_4 \trianglerighteq V_4 \trianglerighteq \langle v \rangle \trianglerighteq \mathbf{1},$$

one for each $v \in V_4$ with $v \neq e$. The factors have orders 2, 3, 2, 2, so they are C_2, C_3, C_2, C_2 (Corollary 1.17), the same for all three, as Theorem 4.10 says.

The derived series. As S_4/A_4 is abelian, $S_4' \leq A_4$. It contains the 3-cycle $[(1\,3), (1\,2)]$ of Example 4.15, and it is normal (Proposition 4.13). Of $\mathbf{1}$, V_4, A_4, only A_4 contains a 3-cycle; so $S_4' = A_4$.

Next, A_4/V_4 has order 3, so it is abelian, and $A_4' \leq V_4$. It contains

$$\begin{aligned}[(1\,2\,3), (1\,2)(3\,4)] &= (1\,2\,3)\,(1\,2)(3\,4)\,(1\,2\,3)^{-1} \cdot (1\,2)(3\,4) \\ &= (1\,4)(2\,3) \cdot (1\,2)(3\,4) = (1\,3)(2\,4),\end{aligned}$$

so it is not $\mathbf{1}$; it is normal in A_4, so $A_4' = V_4$. Finally $V_4' = \mathbf{1}$, as V_4 is abelian. The derived series is

$$S_4 \trianglerighteq A_4 \trianglerighteq V_4 \trianglerighteq \mathbf{1},$$

with factors C_2, C_3 and $V_4 \cong C_2 \times C_2$. It is shorter than a composition series: its last factor is abelian but not simple.

The abelianisation.

$$S_4/S_4' = S_4/A_4 \cong C_2.$$

By the factorisation property in Proposition 4.13, every homomorphism from S_4 to an abelian group factors through the sign.

Not nilpotent. Only e has a class of one element in S_4 (§ 2.6), so $Z(S_4) = \mathbf{1}$, and, as for S_3, every Z_i is $\mathbf{1}$. So S_4 is solvable but not nilpotent.

Self-check. Is A_4 solvable? Is it nilpotent? *Answer:* solvable, with derived series $A_4 \trianglerighteq V_4 \trianglerighteq \mathbf{1}$. Not nilpotent: its classes have 1, 3, 4, 4 elements, so $Z(A_4) = \mathbf{1}$, and every Z_i is $\mathbf{1}$.

Summary.

- *The main results:* the splitting test for classes of A_n (Lemma 4.1) and the simplicity of A_5 (Theorem 4.4); the second isomorphism theorem (Theorem 4.6) and Jordan–Hölder: the composition factors of a finite group are unique up to order and isomorphism (Theorem 4.10); solvability passes to subgroups, quotients and extensions (Theorem 4.17), and S_n is solvable exactly when $n \leq 4$ (Corollary 4.18); p-groups are nilpotent, and nilpotent groups solvable (Theorem 4.21).
- *Handed on:* G', with G/N abelian exactly when $G' \leq N$ (Proposition 4.13), to Lecture 6's one-dimensional representations; solvability and its closure under extensions (Theorem 4.17) to Lecture 9's theorem on solvability of groups of order $p^a q^b$.
- *Next:* Lecture 5 begins the representations: when can a linear action of a group be split into irreducible pieces?

References

Dummit and Foote, *Abstract Algebra*: the second isomorphism theorem, §3.3, Thm 18, p. 97; simple groups and composition series, §3.4, pp. 102–103, Jordan–Hölder, Thm 22, p. 103, its proof outlined in Exercises 9–10, p. 106; solvable groups and extensions, p. 105; A_5 simple, §4.3, Thm 12, p. 128, and the splitting of classes, Exercises 19–20, p. 131; commutators, §5.4, Prop. 7, pp. 169–170; the upper central series, §6.1, p. 190, and p-groups nilpotent, Prop. 2, p. 191; the derived series, Thm 9 and Prop. 10, pp. 195–196. Alperin and Bell, *Groups and Representations*: Jordan–Hölder by two maximal normal subgroups, §10, pp. 91–92. Rotman, *An Introduction to the Theory of Groups*: commutators as here, and G/H abelian iff $G' \leq H$, Thm 2.23, p. 33.